

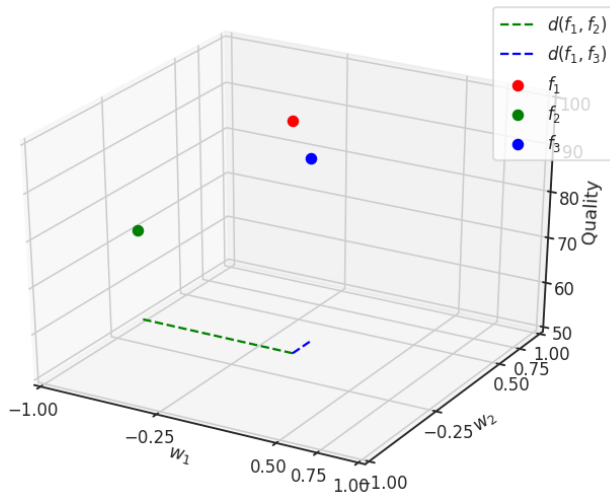
Projection to latent space

MIPT

2025

Motivation

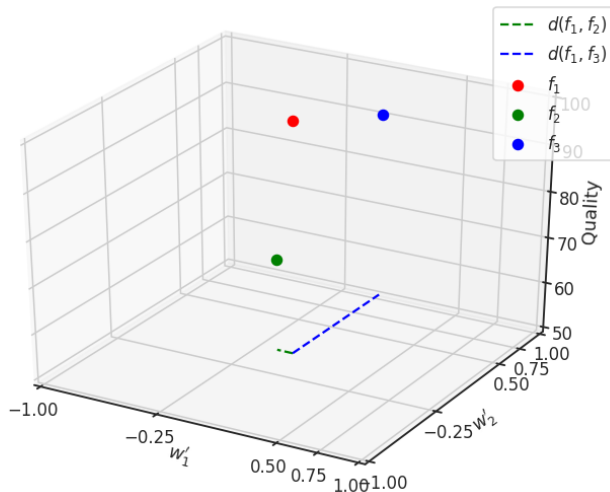
Which model is closer to f_1 ?



Motivation

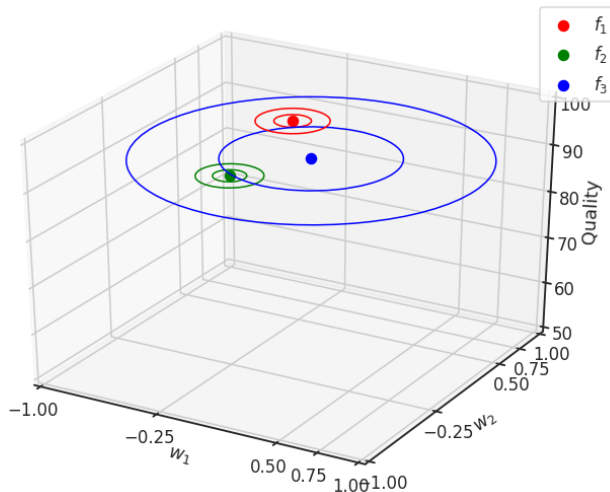
Which model is closer to f_1 ?

Metric change $\approx$ coordinate change. Different metrics represent different model space properties.



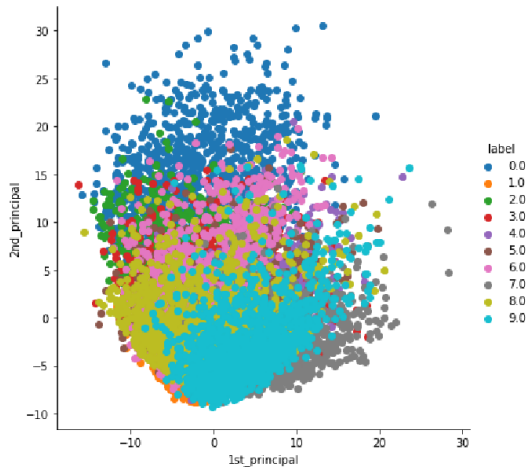
Motivation

Which model is closer to f_1 ?



Principal component analysis

$$\mathbf{W} = \arg \max \text{Var}(\mathbf{X}\mathbf{W})$$

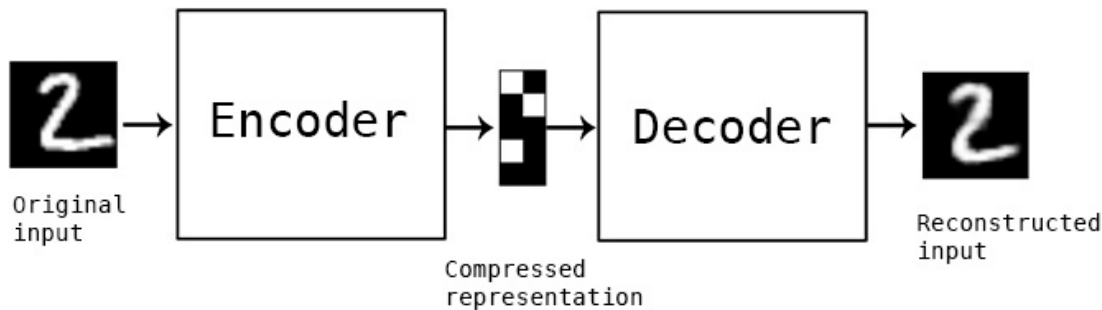


Autoencoder

Autoencoder is a model of dimension reduction:

$$\mathbf{H} = \sigma(\mathbf{W}_e \mathbf{X}),$$

$$\|\sigma(\mathbf{W}_d \mathbf{H}) - \mathbf{X}\|_2^2 \rightarrow \min.$$



Multiple spaces

Given two spaces: **X**, **Y**.

How we can build a shared latent space between them?

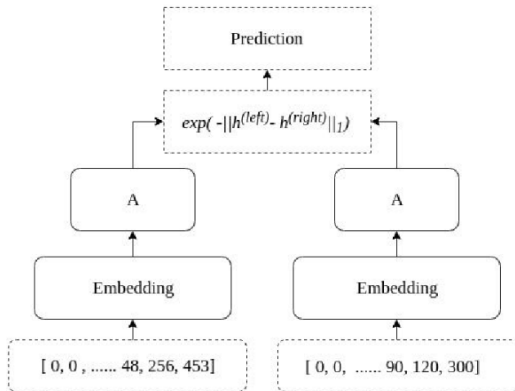
Multiple spaces

Given two spaces: **X**, **Y**.

How we can build a shared latent space between them?

Naive method: $\|\mathbf{f}(\mathbf{x}) - \mathbf{g}(\mathbf{y})\|_2^2 \rightarrow \min$ does not work.

Siamese networks



Metric learning

$$D(\mathbf{x}_1, \mathbf{x}_2) = \sqrt{(\mathbf{x}_1 - \mathbf{x}_2)^\top \mathbf{M}(\mathbf{x}_1 - \mathbf{x}_2)}$$

Triplet loss

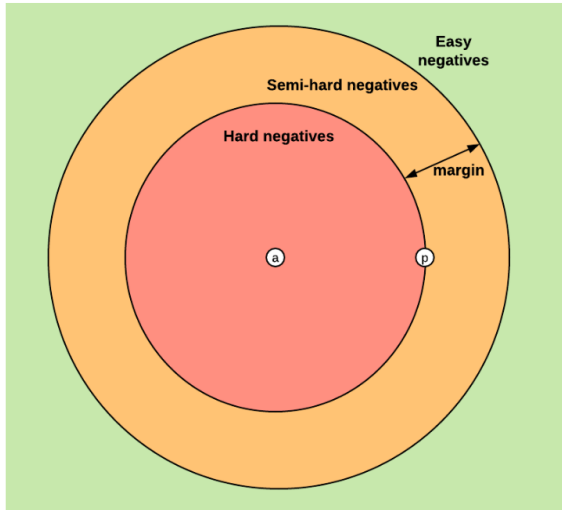
The loss function for each sample in the mini-batch is:

$$L(a, p, n) = \max\{d(a_i, p_i) - d(a_i, n_i) + \text{margin}, 0\}$$

where

$$d(x_i, y_i) = \|\mathbf{x}_i - \mathbf{y}_i\|_p$$

Triplet loss



Bayesian representation learning with oracle constraints

$$p(t_{i,j,l}) = \int_{\mathbf{z}} p(t_{i,j,l} | z_i, z_j, z_l) p(\mathbf{z}_i) p(\mathbf{z}_j) p(\mathbf{z}_l) d\mathbf{z}_i d\mathbf{z}_j d\mathbf{z}_l,$$

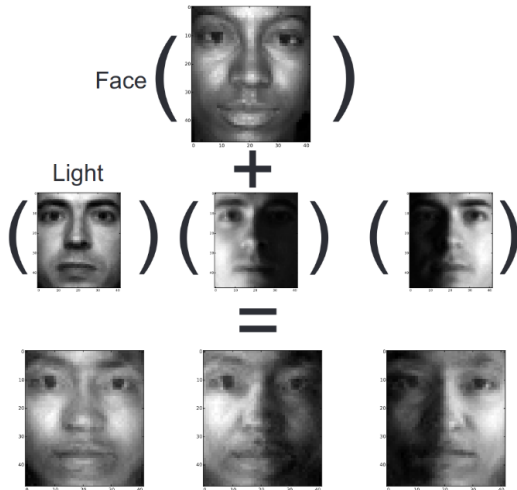
this gives the following likelihood:

$$p(t_{i,j,l}) = \text{Ber}(t_{i,j,l}) = \frac{e^{-D_{i,j}}}{e^{-D_{i,j}} + e^{-D_{i,l}}}$$

with

$$D_{a,b} = \sum_{h=1}^H D_{a,b}^h = - \sum_{h=1}^H \left[\text{JS} \left(p(\mathbf{z}_a^h) || p(\mathbf{z}_b^h) \right) \right].$$

Bayesian representation learning with oracle constraints



Variational learning across domains with triplet information

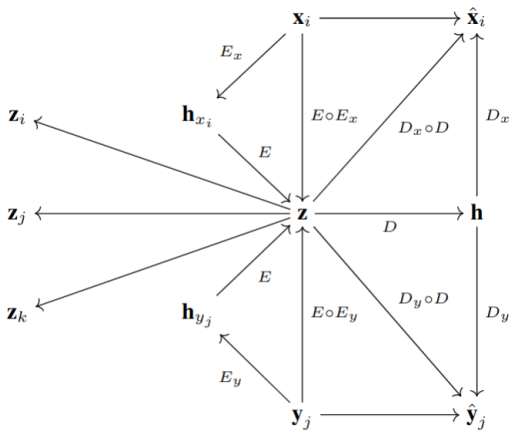


Figure 1: VBTA generative process

Variational learning across domains with triplet information

$$\begin{aligned}
 \mathcal{L}_{VBTA} &= \mathbb{E}_{q_{\phi_x}(\mathbf{z}_x|\mathbf{x})} \log \frac{p_{\theta_x}(\mathbf{x}, \mathbf{y}, \mathbf{t}, \mathbf{z}_x)}{q_{\phi_x}(\mathbf{z}_x|\mathbf{x})} + \mathbb{E}_{q_{\phi_y}(\mathbf{z}_y|\mathbf{y})} \log \frac{p_{\theta_y}(\mathbf{x}, \mathbf{y}, \mathbf{t}, \mathbf{z}_y)}{q_{\phi_y}(\mathbf{z}_y|\mathbf{y})} = \\
 &= \underbrace{- \left[KL(q_{\phi_x}(\mathbf{z}_x|\mathbf{x}) \parallel p_{\theta_x}(\mathbf{z}_x)) + KL(q_{\phi_y}(\mathbf{z}_y|\mathbf{y}) \parallel p_{\theta_y}(\mathbf{z}_y)) \right]}_{\text{Penalty}} + \\
 &\quad + \underbrace{\left[\mathbb{E}_{q_{\phi_x}(\mathbf{z}_x|\mathbf{x})} [\log p_{\theta_x}(\mathbf{x}|\mathbf{z}_x)] + \mathbb{E}_{q_{\phi_y}(\mathbf{z}_y|\mathbf{y})} [\log p_{\theta_y}(\mathbf{y}|\mathbf{z}_y)] \right]}_{\text{Reconstruction}} + \\
 &\quad + \underbrace{\left[\mathbb{E}_{q_{\phi_x}(\mathbf{z}_x|\mathbf{x})} [\log p_{\theta_x}(\mathbf{y}|\mathbf{z}_x)] + \mathbb{E}_{q_{\phi_y}(\mathbf{z}_y|\mathbf{y})} [\log p_{\theta_y}(\mathbf{x}|\mathbf{z}_y)] \right]}_{\text{Cycle-consistency}} + \\
 &\quad + \underbrace{\mathbb{E}_{q_{\phi_x}(\mathbf{z}_x|\mathbf{x})} [\log p(\mathbf{t}|\mathbf{z}_x)] + \mathbb{E}_{q_{\phi_y}(\mathbf{z}_y|\mathbf{y})} [\log p(\mathbf{t}|\mathbf{z}_y)]}_{\text{Triplet likelihood}}
 \end{aligned}$$

How we can embed models into (probabilistic) vector space?

Definition and properties

Given a parameter space $\mathbf{w}$.

A distance function d is a function, defined on the pair of distributions $p_1, p_2 \rightarrow \mathbb{R}_+$.

Probable Properties

- Metric axioms
 - ▶ $d(p_1, p_1) = 0$
 - ▶ $d(p_1, p_2) = d(p_2, p_1)$
 - ▶ $d(p_1, p_2) \leq d(p_1, p_3) + d(p_3, p_2)$
- (Aduenko, 2017)
 - ▶ $d \in [0, 1]$
 - ▶ d is defined in case of different support for p_1, p_2
 - ▶ d is nearly zero, if p_2 is a low-informative distribution
- Performance criteria
 - ▶ Tractable
 - ▶ Easy to compute

Total variation

For two probability measures P_1, P_2 on the set $\mathfrak{A}$

$$TV = \sup_{a \in \mathfrak{A}} |P_1(a) - P_2(a)|$$

Properties:

- $0 \leq TV \leq 1$
- TV is a metric
- $TV = 0 \iff P_1 = P_2$
- Scheffe lemma: for differentiable distributions with PDF f_i defined on $\mathbb{R}^d$:

$$TV = \frac{1}{2} \int |f_1(\mathbf{x}) - f_2(\mathbf{x})| d\mathbf{x} = \frac{1}{2} \|f_1 - f_2\|_1.$$

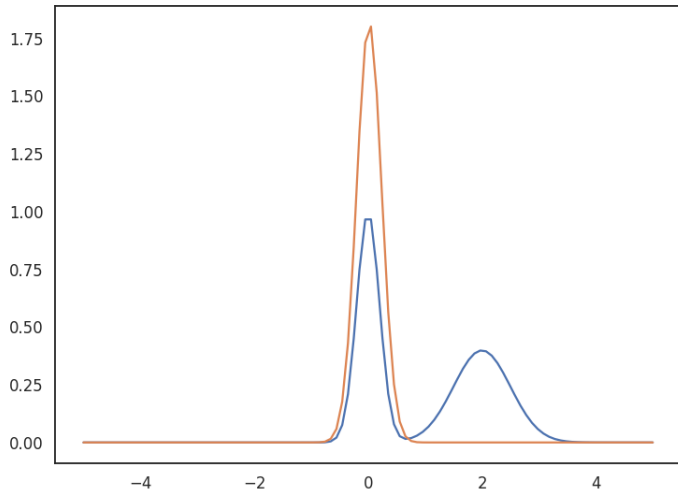
- $TV(\prod_i P_1^i, \prod_i P_2^i) \leq \sum_i TV(P_1^i, P_2^i)$
- Corresponds to statistics in KS-test

Hellinger distance

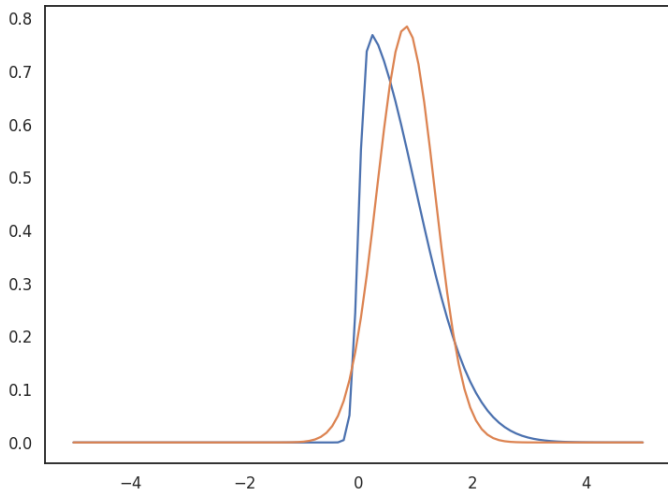
$$H = \sqrt{\int (f_1(\mathbf{x}) - f_2(\mathbf{x}))^2 d\mathbf{x}} = \|\sqrt{f_1} - \sqrt{f_2}\|_2$$

- $0 \leq H \leq 2$
- H is metric
- $H = 0 \iff P_1 = P_2$
- $H^2(\prod_i P_1^i, \prod_i P_2^i) \leq \sum_i H^2(P_1^i, P_2^i)$
- $1 - H^2 = \int \sqrt{f_1(\mathbf{x})f_2(\mathbf{x})} d\mathbf{x}$

Hellinger distance: example



Hellinger distance: example



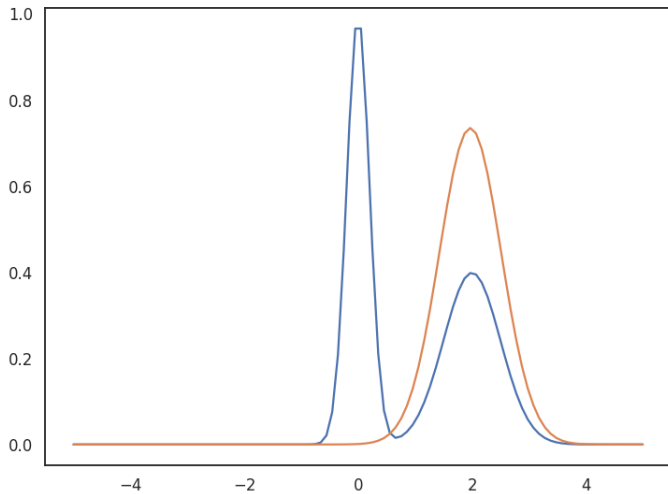
KL divergence

$$KL(P_1, P_2) = \int \log \frac{f_1(x)}{f_2(x)} f_1(x) dx$$

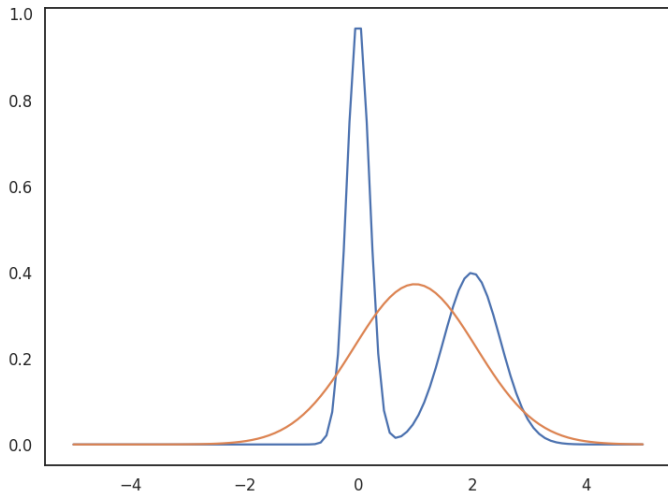
- $KL \geq 0$
- KL is not a metric: not a symmetric
- KL is not a metric: does not respect triangle inequality
- $KL = 0 \iff P_1 = P_2$
- $KL(\prod_i P_1^i, \prod_i P_2^i) = \sum_i KL(P_1^i, P_2^i)$
- If we have a dependence between 2 random values $\mathbf{w}, \gamma$, then

$$KL(p_1(\mathbf{w}, \gamma), p_2(\mathbf{w}, \gamma)) = KL(p_1(\mathbf{w}), p_2(\mathbf{w})) + \int_{\mathbf{w}} p_1(\mathbf{w}) \int_{\gamma} \log \frac{p_1(\gamma|\mathbf{w})}{p_2(\gamma|\mathbf{w})} p_1(\gamma|\mathbf{w}) d\gamma d\mathbf{w}$$

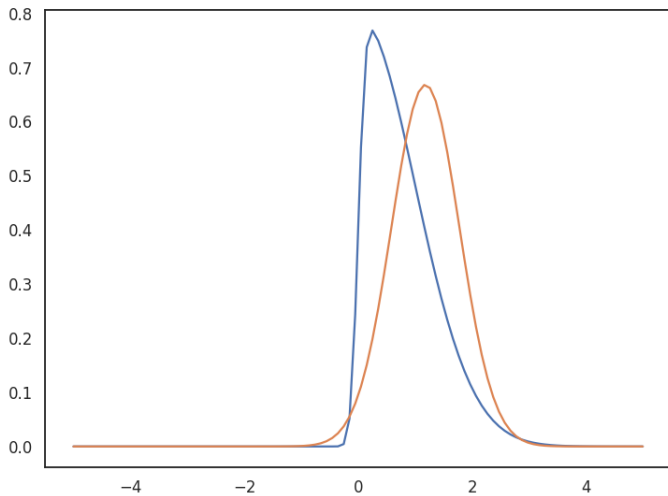
KL: example



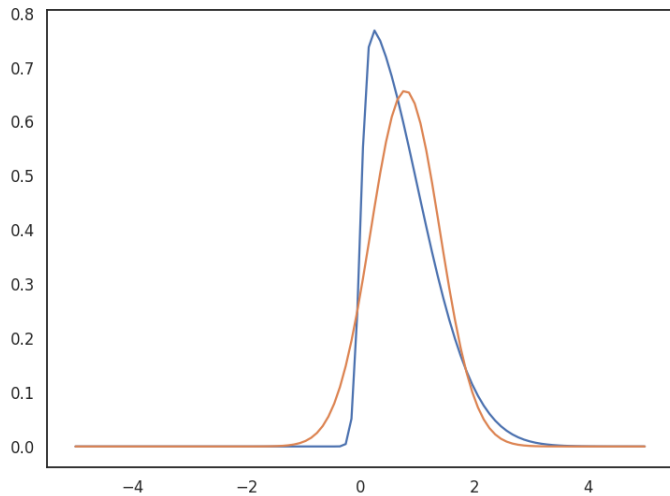
KL: example



KL: example



KL: example



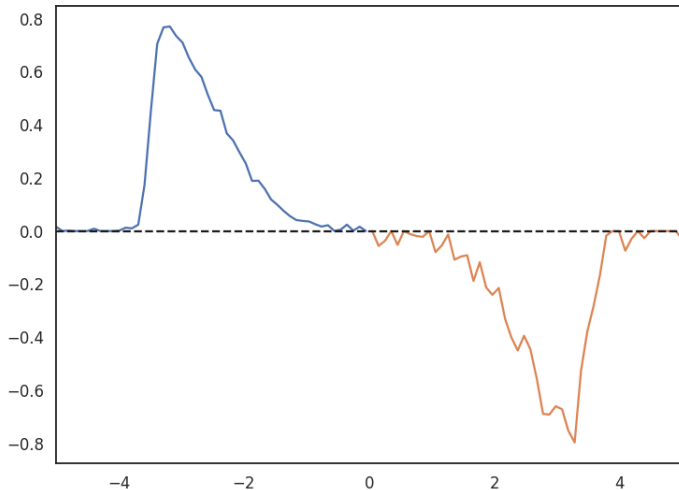
JS

$$JS(P_1, P_2) = \frac{1}{2}KL\left(P_1 \middle| \frac{1}{2}P_1 + \frac{1}{2}P_2\right) + \frac{1}{2}KL\left(P_2 \middle| \frac{1}{2}P_1 + \frac{1}{2}P_2\right)$$

- $0 \leq JS \leq 1$
- $\sqrt{JS}$ is a metric
- $JS = 0 \iff P_1 = P_2$

Wasserstein distance: motivation

Gaspard Monge: how to move sand into hole in a cheapest way?



Wasserstein distance: discrete problem

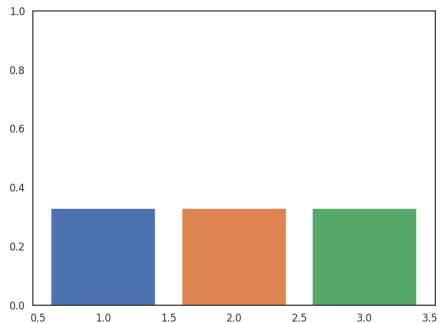
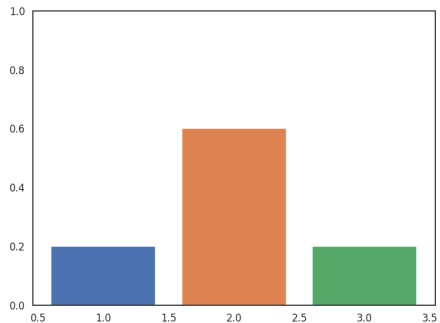
Given two discrete probability measures $p_1(\mathbf{w}_i^1), i \in \{1, \dots, n_1\}$, $p_2(\mathbf{w}_j^2), j \in \{1, \dots, n_2\}$.

Given a cost matrix $\mathbf{C}$: $c_{ij} \in \mathbb{R}_+$.

We need to find a mapping induced by matrix t_{ij} that:

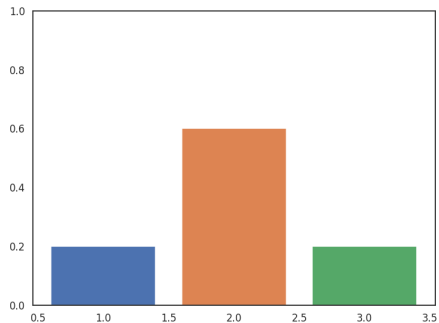
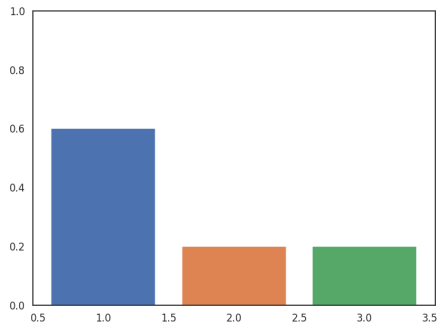
- $\sum_i t_{ij} = p_2(\mathbf{w}_j^2), \sum_j t_{ij} p_2(\mathbf{w}_i^1)$
- $\sum_i \sum_j c_{ij} t_{ij} \rightarrow \min.$

Discrete problem: example



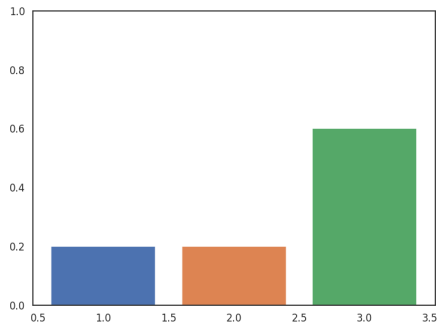
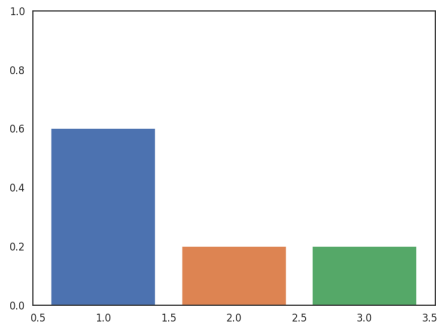
Cost: 0.4

Discrete problem: example



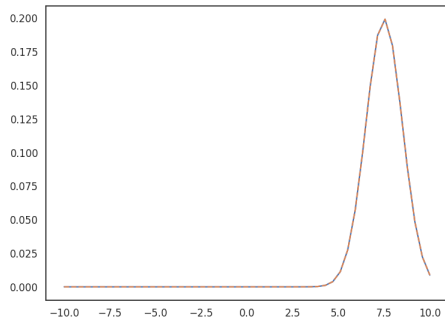
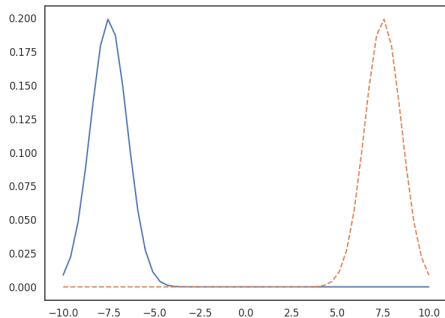
Cost: 0.4

Discrete problem: example

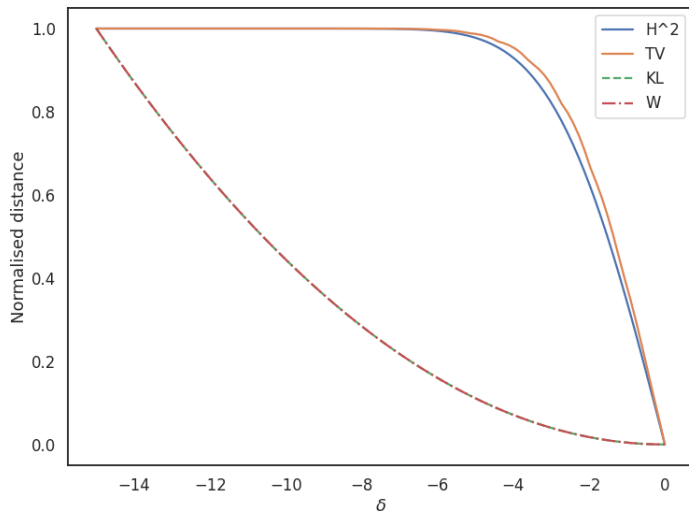


Cost: 0.8

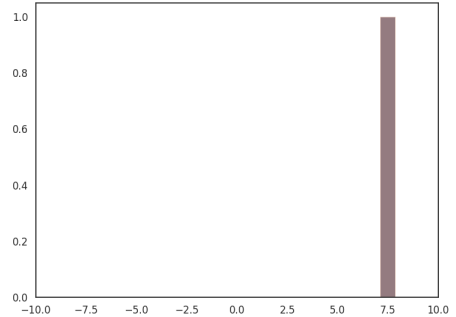
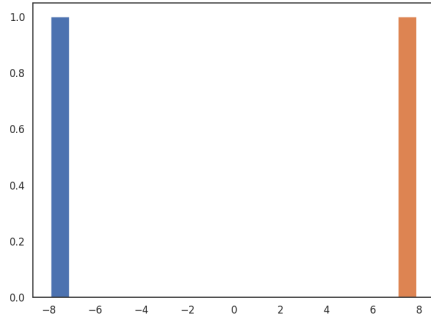
Distance between peaks: example



Distance between peaks: example



Distance between peaks: example



Distance between peaks: example

$$TV = 0$$

$$H = 0$$

$$KL = \begin{cases} 0, & \delta = 0 \\ \infty, & \text{otherwise} \end{cases}$$

$$JS = \begin{cases} 0, & \delta = 0 \\ \log 2, & \text{otherwise} \end{cases}$$

$$W = |\delta|.$$

Conclusion: W-distance has good properties to work with different support sets.

What do we want from these embeddings?

Differentiable Neural Architecture Search in Equivalent Space with Exploration Enhancement

- Structure representation: graph supervised encoder
- Structure optimization: DARTS + exploration

Table 1: Comparison results with state-of-the-art NAS approaches on NAS-Bench-201.

Method	CIFAR-10		CIFAR-100		ImageNet-16-120	
	Valid(%)	Test(%)	Valid(%)	Test(%)	Valid(%)	Test(%)
ENAS	37.51 $\pm$ 3.19	53.89 $\pm$ 0.58	13.37 $\pm$ 2.35	13.96 $\pm$ 2.33	15.06 $\pm$ 1.95	14.84 $\pm$ 2.10
RandomNAS*	85.63 $\pm$ 0.44	88.58 $\pm$ 0.21	60.99 $\pm$ 2.79	61.45 $\pm$ 2.24	31.63 $\pm$ 2.15	31.37 $\pm$ 2.51
DARTS (1st)	39.77 $\pm$ 0.00	54.30 $\pm$ 0.00	15.03 $\pm$ 0.00	15.61 $\pm$ 0.00	16.43 $\pm$ 0.00	16.32 $\pm$ 0.00
DARTS (2nd)	39.77 $\pm$ 0.00	54.30 $\pm$ 0.00	15.03 $\pm$ 0.00	15.61 $\pm$ 0.00	16.43 $\pm$ 0.00	16.32 $\pm$ 0.00
SETN	84.04 $\pm$ 0.28	87.64 $\pm$ 0.00	58.86 $\pm$ 0.06	59.05 $\pm$ 0.24	33.06 $\pm$ 0.02	32.52 $\pm$ 0.21
NAO*	82.04 $\pm$ 0.21	85.74 $\pm$ 0.31	56.36 $\pm$ 3.14	59.64 $\pm$ 2.24	30.14 $\pm$ 2.02	31.35 $\pm$ 2.21
GDAS*	90.03 $\pm$ 0.13	93.37 $\pm$ 0.42	70.79 $\pm$ 0.83	70.35 $\pm$ 0.80	40.90 $\pm$ 0.33	41.11 $\pm$ 0.13
E ² NAS	90.94$\pm$0.83	93.89$\pm$0.47	71.83$\pm$1.84	72.05$\pm$1.58	45.44$\pm$1.24	45.77$\pm$1.00

Does Unsupervised Architecture Representation Learning Help Neural Architecture Search?

- Structure representation: graph VAE
- Optimization: unsupervised for encoding models, then RL+BO

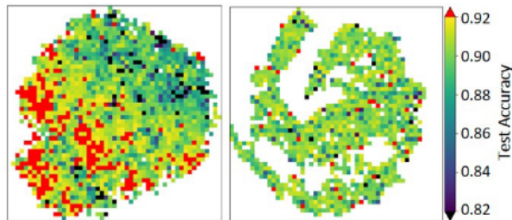
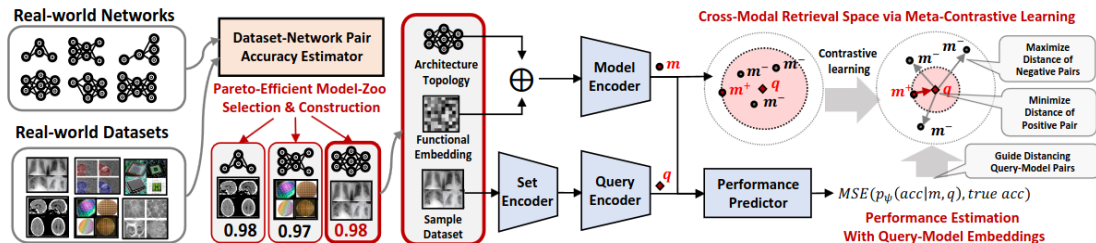


Figure 4: Latent space 2D visualization [65] comparison between *arch2vec* (left) and supervised architecture representation learning (right) on NAS-Bench-101. Color encodes test accuracy. We randomly sample 10,000 points and average the accuracy in each small area.

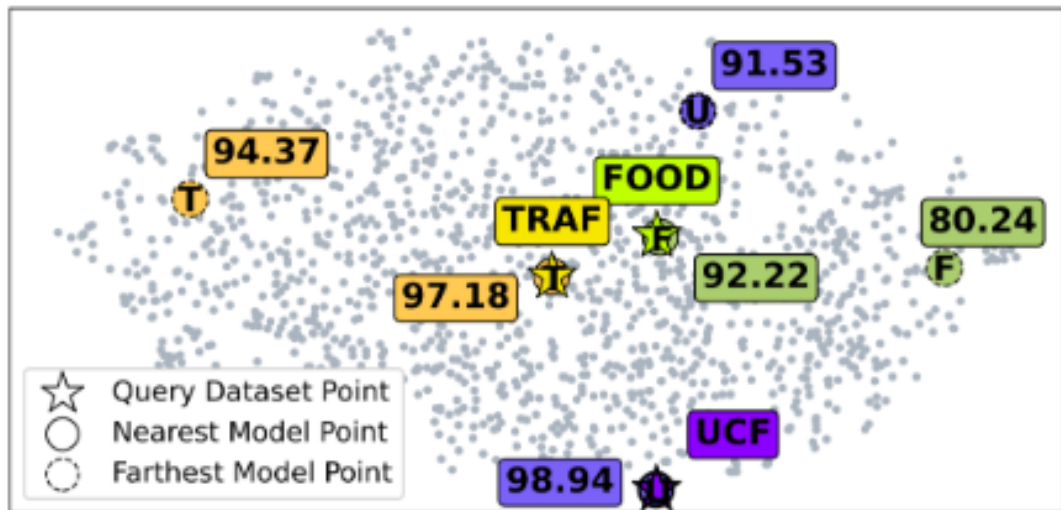
Task-Adaptive Neural Network Search with Meta-Contrastive Learning



Task-Adaptive Neural Network Search with Meta-Contrastive Learning

Target Dataset	Method	# Epochs	FLOPs (M)	Params (M)	Search Time (GPU sec)	Training Time (GPU sec)	Speed Up	Accuracy (%)
Averaged Performance	MobileNetV3 [26]	50	132.94	4.00	-	257.78±09.77	1.00×	94.20±0.70
	PC-DARTS [65]	500	566.55	3.54	1100.37±22.20	5721.13±793.71	0.04×	79.22±1.69
	DrNAS [10]	500	623.43	4.12	1501.75±43.92	5659.77±403.62	0.04×	84.06±0.97
	FBNet-A [60]	50	246.69	4.3	-	293.42±57.45	0.88×	93.00±1.95
	OFA [8]	50	148.76	6.74	121.90±0.00	226.58±03.13	0.74×	93.89±0.84
	MetaD2A [31]	50	512.67	6.56	2.59±0.13	345.39±28.36	0.74×	95.24±1.14
	TANS (Ours)	10	181.74	5.51	0.002±0.00	40.19±03.06	-	95.17±2.20
	TANS (Ours)	50	181.74	5.51	0.002±0.00	200.93±11.01	1.28×	96.28±0.30
Colorectal Histology Dataset (Easy)	MobileNetV3 [26]	50	132.94	4.00	-	577.18±04.15	1.00×	96.23±0.07
	PC-DARTS [65]	500	534.64	4.02	2062.42±49.14	12124.18±1051.16	0.04×	96.17±0.68
	DrNAS [10]	500	614.23	4.12	4183.20±188.60	11355.18±1352.62	0.04×	97.51±0.13
	FBNet-A [60]	50	215.45	4.3	-	696.00±295.19	0.83×	95.43±0.57
	OFA [8]	50	134.85	6.74	121.90±0.00	537.61±03.52	0.88×	96.40±0.52
	MetaD2A [31]	50	506.88	5.93	2.58±0.12	784.45±79.32	0.73×	96.57±0.56
	TANS (Ours)	10	171.74	4.95	0.001±0.00	98.56±04.24	-	96.87±0.21
	TANS (Ours)	50	171.74	4.95	0.001±0.00	492.81±21.19	1.17×	97.67±0.05
Food Classification Dataset (Hard)	MobileNetV3 [26]	50	132.94	4.00	-	235.57±07.57	1.00×	87.52±0.78
	PC-DARTS [65]	500	567.85	3.62	1018.49±6.31	6323.40±938.83	0.03×	55.42±2.46
	DrNAS [10]	500	632.67	4.12	1276.38±0.00	5079.89±161.05	0.04×	61.45±0.68
	FBNet-A [60]	50	251.29	4.3	-	251.24±3.31	0.94×	84.33±1.41
	OFA [8]	50	152.34	6.74	121.90±0.00	190.86±03.48	0.75×	87.43±0.59
	MetaD2A [31]	50	521.11	8.23	2.60±0.23	324.62±34.97	0.72×	83.77±0.38
	TANS (Ours)	10	152.34	6.74	0.001±0.00	190.86±03.48	-	87.43±0.59

Task-Adaptive Neural Network Search with Meta-Contrastive Learning



Recap: PCA (1 component)

Given a dataset $\mathbf{X}$. We want to find a weight vector $\mathbf{p}$, $\|\mathbf{p}\|^2 = 1$ that the linear combination

$$\sum_{\mathbf{x}} \sum_{j=1}^d p_j \mathbf{x}_j \rightarrow \max$$

Can we repeat the same technique for the functions?

PCA for functional spaces

Given a set of functions $\mathbf{X}(s)$ in a Banach space. We want to find a weight vector $\mathbf{p}$, $\|\mathbf{p}\|^2 = 1$ that the linear combination

$$\sum_{\mathbf{x}} \sum_{j=1}^d \int (\mathbf{p}(s)\mathbf{x}(s)ds)^2 \rightarrow \max$$

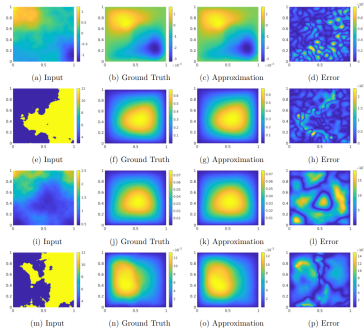
- How the $\mathbf{X}(s)$ and $\mathbf{X}$ are connected?
- What are «good» functions for use this method?

Operator learning

Example 1 We want to approximate a differential operator $\mathcal{P}_x$.

$$(\mathcal{P}_x y)(s) = 0 \forall s \in D,$$

where x and y are **functions** from Banach space, and $D \subset \mathbb{R}^d$.



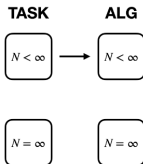
Operator learning



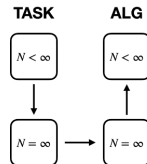
(A) Same images at different resolutions



(B) Different resolutions as vectors and (bottom right) as a function



(A) Directly design algorithm at fixed resolution N



(B) Design algorithm at limit of infinite resolution

Example 2

PCA-net

- ① Learn PCA for input (the input functions are discretized, so we can use a simple Euclidean basis for learning PCA)
- ② Learn PCA for output
- ③ Learn NN to transform input to output

$$\begin{array}{ccccc} \mathcal{U} & \xrightarrow{F_{\mathcal{U}}} & \mathbb{R}^{d_{\mathcal{U}}} & \xrightarrow{G_{\mathcal{U}}} & \mathcal{U} \\ \Psi^{\dagger} \downarrow & & \varphi \downarrow & & \Psi^{\dagger} \downarrow \\ \mathcal{V} & \xrightarrow{F_{\mathcal{V}}} & \mathbb{R}^{d_{\mathcal{V}}} & \xrightarrow{G_{\mathcal{V}}} & \mathcal{V} \end{array}$$

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- Final classes: next week
- For each project we expect a short report with answers to all the reviewer comments:
 - ▶ What was fixed
 - ▶ What wasn't fixed (any why)
 - ▶ Blog-post status: (published, not published, when will be published)
 - ▶ Do you want to publish a library (at least we can try to publish it)
 - ▶ **Optional:** some demo (if you have some updates and want to share with other students)
- For Multi-task learning: some comments will be ready on Friday